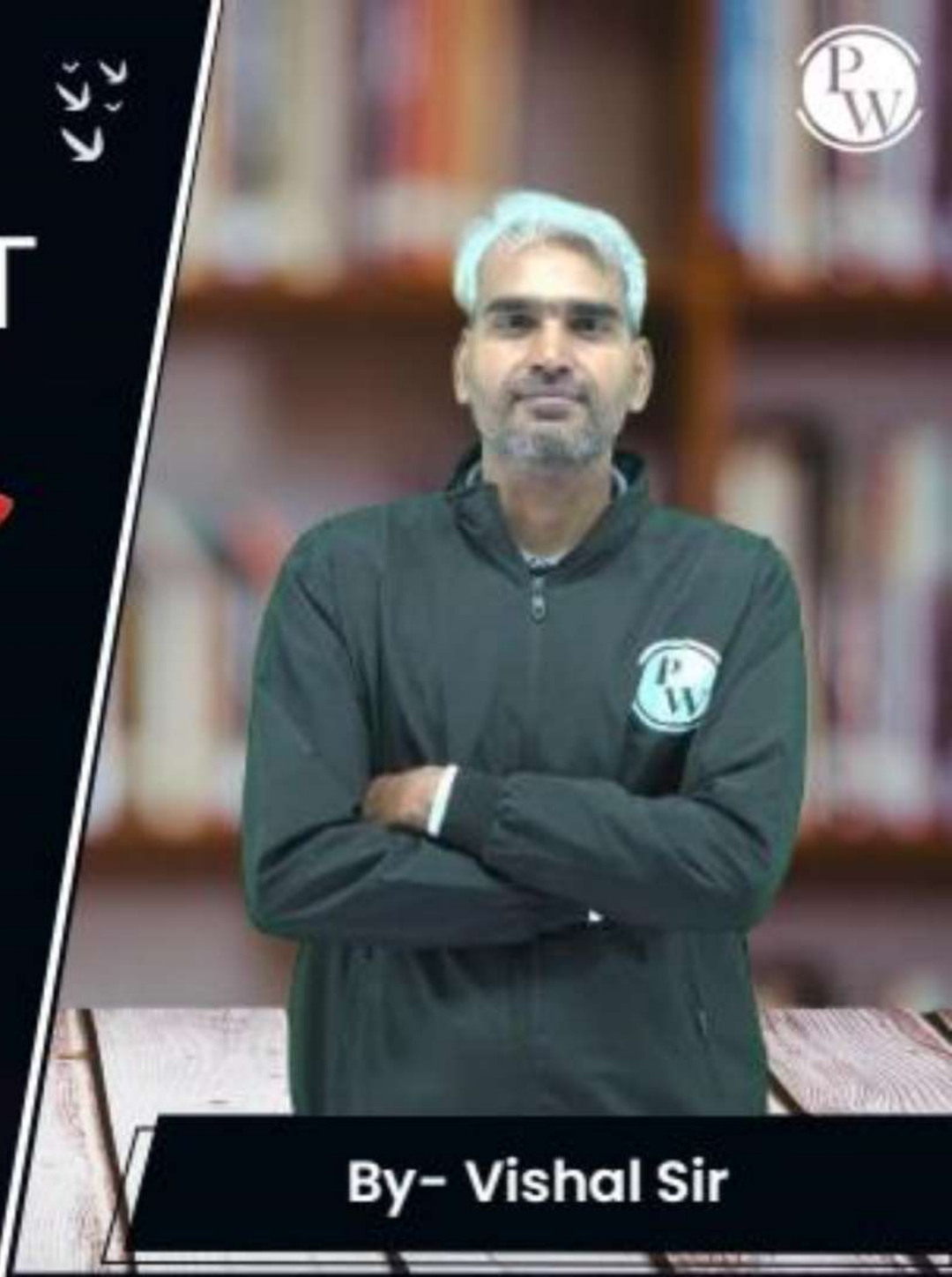
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 01



By- Vishal Sir

Topics to be Covered



- Topic: Syllabus
- Topic: Introduction to DBMS
- Topic: Relational Database Model

Slide



Topic : Syllabus

- Relational Model & Normal Forms
- Query Languages
- Transactions and Concurrency Control
- File Organization & Indexing
- ER model

Relational Model
Functional dependency
Closure of attribute set
Different types of keys
Minimal Cover
Relation b/w FD sets
Normalization
Lossless Join decomposition
Dependency preserving decomposition
Normal forms
- 1NF
- 2NF
- 3NF
- BCNF
- 4NF



Topic : Syllabus

- Relational Model & Normal Forms
- Query Languages
  - Relational Algebra
  - SQL
  - Tuple Relational Calculus

  ← Most Imp
- Transactions and Concurrency Control
- File Organization & Indexing
- ER model



Topic : Syllabus

- Relational Model & Normal Forms
- Query Languages
- Transactions and Concurrency Control
- File Organization & Indexing
- ER model

Transactions & Schedules
ACID Property
Recoverable Schedule & Serializable Schedule
Concurrency Control Protocols
lock based Protocols
Time stamp based Protocol



Topic : Syllabus

- Relational Model & Normal Forms
- Query Languages
- Transactions and Concurrency Control
- File Organization & Indexing
- ER model

Classification of Indexing
Primary, Secondary & Clustering Index
Multi-level Index
Dynamic Multilevel Index
IMP → B Tree, B+ Tree



Topic : Syllabus

- Relational Model & Normal Forms
- Query Languages
- Transactions and Concurrency Control
- File Organization & Indexing
- ER model
  - Entity Relationship Model
  - Conversion from ER Model into Relational Table



Topic : Introduction to DBMS

Data (Raw data) → May not provide any specific information
eg: 15, 23, 5, 30

Information {Meaningful data}
Let 15, 23, 5, 30 represent the number of students enrolled in different courses.

Database {Organized Collection of related information}

Data can be organized in different ways; ∴ There are different types of databases
(In this course we will discuss about Relational database)

DBMS → {DBMS is a software that allow us to manage & access the database efficiently}

Course	No. of students enrolled	fee
Parakram-A	15	
Vijay	23	
DS4AI	5	
Shreshtha	30	

Database

{ Relational }

↳ Because stored in the form of table



Topic : Relational database

* In Relational database information is stored in the form of table.

  └ Table is a collection of Rows & Columns



Topic : Relational database

Attribute / field

eg:

Student Database

* Degree/Arity = 3
* Cardinality = 4
* Relational Schema
  = Student (Sid, Sname, Mob_No)

Sid	Sname	Mob_No
S_1	A	123
S_2	B	456
S_3	A	222
S_4	C	NULL

3 attributes in the Student table
Sid, Sname, Mob_No.

let, Name of the Relation is
"Student"



Topic : Relational database

① Arity / Degree : No. of attributes/fields in the relational table defines the degree/Arity of that Relation.

② Tuple (or Record) : Each row in relational table is called tuple / Record. {There are 4 tuples in the above Relation}

* **Cardinality** : Number of tuples in a relational table defines the Cardinality of that relation

* **Relational Schema**:- It is use to provide the abstract detail of the relational table

eg: Student (Sid, Sname, Mob_No)

This Relational schema tells us, that there is a relation in the database with name "Student" which Consist of three attributes Sid, Sname & Mob_No.

Relational Instance :

If data exists in the relational table, then that set of data is called relational instance of that time.

In the above eg:

S_1	A	123
S_2	B	456
S_3	A	222
S_4	C	NULL

is the relational instance of current time

- Relational instance of relation may change with time, because we may perform instetion, deletion or updation into the Relation



2 mins Summary

Topic Introduction to DBMS

Topic Relational database

Slide

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THANK - YOU

Slide